IGCSE · Cambridge (CIE) · Maths

50 mins 11 questions

Calculator Questions

Simultaneous Equations

Linear Simultaneous Equations / Quadratic Simultaneous Equations

Medium (5 questions)	/15
Hard (3 questions)	/17
Very Hard (3 questions)	/18
Total Marks	**/50**

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Medium Questions

1 Solve the simultaneous equations.

You must show all your working.

$$\frac{1}{2}x - 3y = 9$$
$$5x + y = 28$$

(3 marks)

2 Solve the simultaneous equations

$$2x + 7y = 17$$
$$5x + 3y = -1$$

Show clear algebraic working.

x = ...

y = ...

(4 marks)

3 Solve the simultaneous equations

$$5a + 2c = 10$$
$$2a - 4c = 7$$

Show clear algebraic working.

a = ...

c = ...

(3 marks)

4 Solve the simultaneous equations

$$7x - 2y = 34$$
$$3x + 5y = -3$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(1 mark)

5 Solve the simultaneous equations.
You must show all your working.

$$2x + 3y = -12$$
$$5x + 2y = 14$$

$x =$ ....
$y =$ ....

(4 marks)

Hard Questions

1 Solve the simultaneous equations.
You must show all your working.

$$x = 7 - 3y$$
$$x^2 - y^2 = 39$$

$x =$ .... $y =$ ....
$x =$ .... $y =$ ....

(6 marks)

2 Solve the simultaneous equations

$$x^2 + y^2 = 9$$
$$x + y = 2$$

You must show all your working.

Give your answers correct to 2 decimal places.

(6 marks)

3 Solve the equations

$$x^2 + y^2 = 36$$
$$x = 2y + 6$$

You must show all your working.

(5 marks)

Very Hard Questions

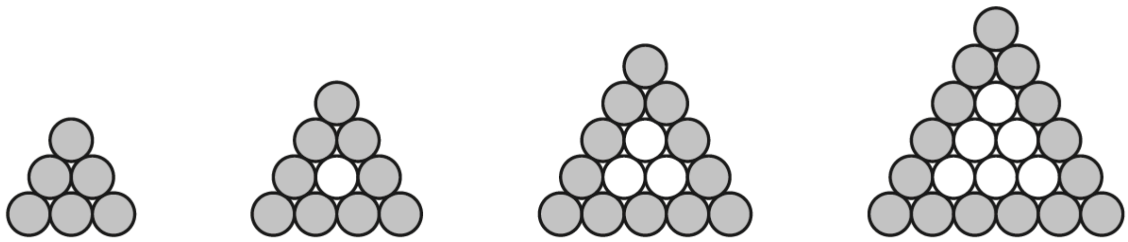
1 Solve the simultaneous equations.
You must show all your working.

$$y = 4 - x$$
$$x^2 + 2y^2 = 67$$

$x =$ .... , $y =$ ....
$x =$ .... , $y =$ ....

(6 marks)

2 Marco is making patterns with grey and white circular mats.



Pattern 1 Pattern 2 Pattern 3 Pattern 4

The patterns form a sequence. Marco makes a table to show some information about the patterns.

Pattern number	1	2	3	4
Number of grey mats	6	9	12	15
Total number of mats	6	10	15	21

Marco needs a total of 6 mats to make the first pattern.
He needs a total of 16 mats to make the first two patterns.
He needs a total of $\frac{1}{6}n^3 + an^2 + bn$ mats to make the first n patterns.

Find the value of a and the value of b.

$a =$ ....
$b =$ ....

(6 marks)

3 The straight line $y = 2x + 1$ intersects the curve $y = x^2 + 3x - 4$ at the points *A* and *B*.

Use an algebraic method to find the coordinates of *A* and *B*.
Give your answers correct to 2 decimal places.

You must show all your working.

A (.................... , ...)
B (.................... , ...)
(6 marks)